# Laboratory screening of sorghum lines for incompatibility, a post-attachment resistance mechanism to the parasitic weed *Striga hermonthica*

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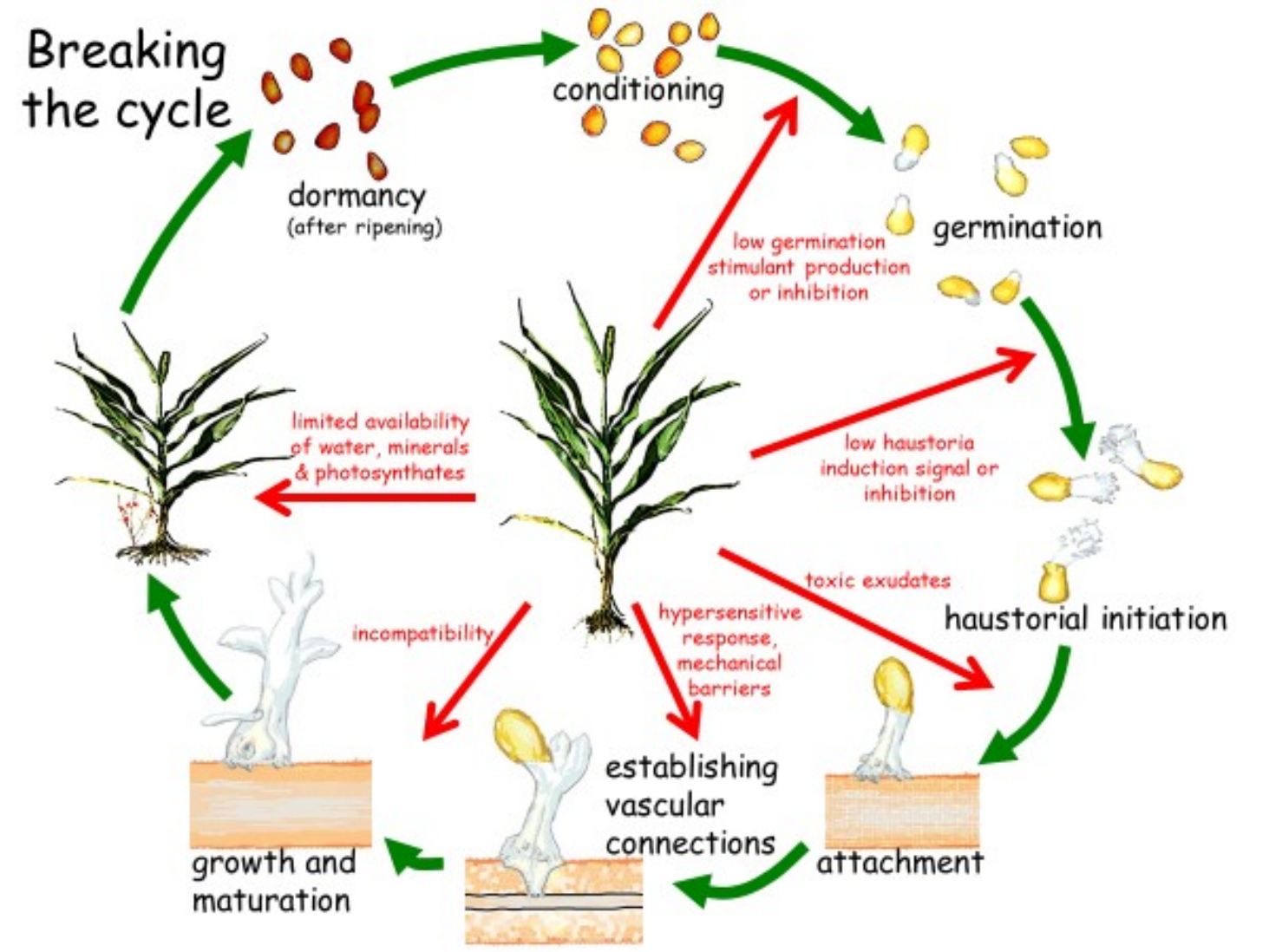
# What is *Striga hermonthica?*

- *Striga* is an obligate root parasitic weed commonly referred to as a witchweed
- *S. hermonthica* infests maize, sorghum, upland rice and millet, mainly in sub-Saharan Africa
- *Striga* perturbs its hosts' nutrient and water relations with severe effects on yield under heavy infestations
- *Striga* is particularly hard on substance farmers who do not use fertilizer and irrigation

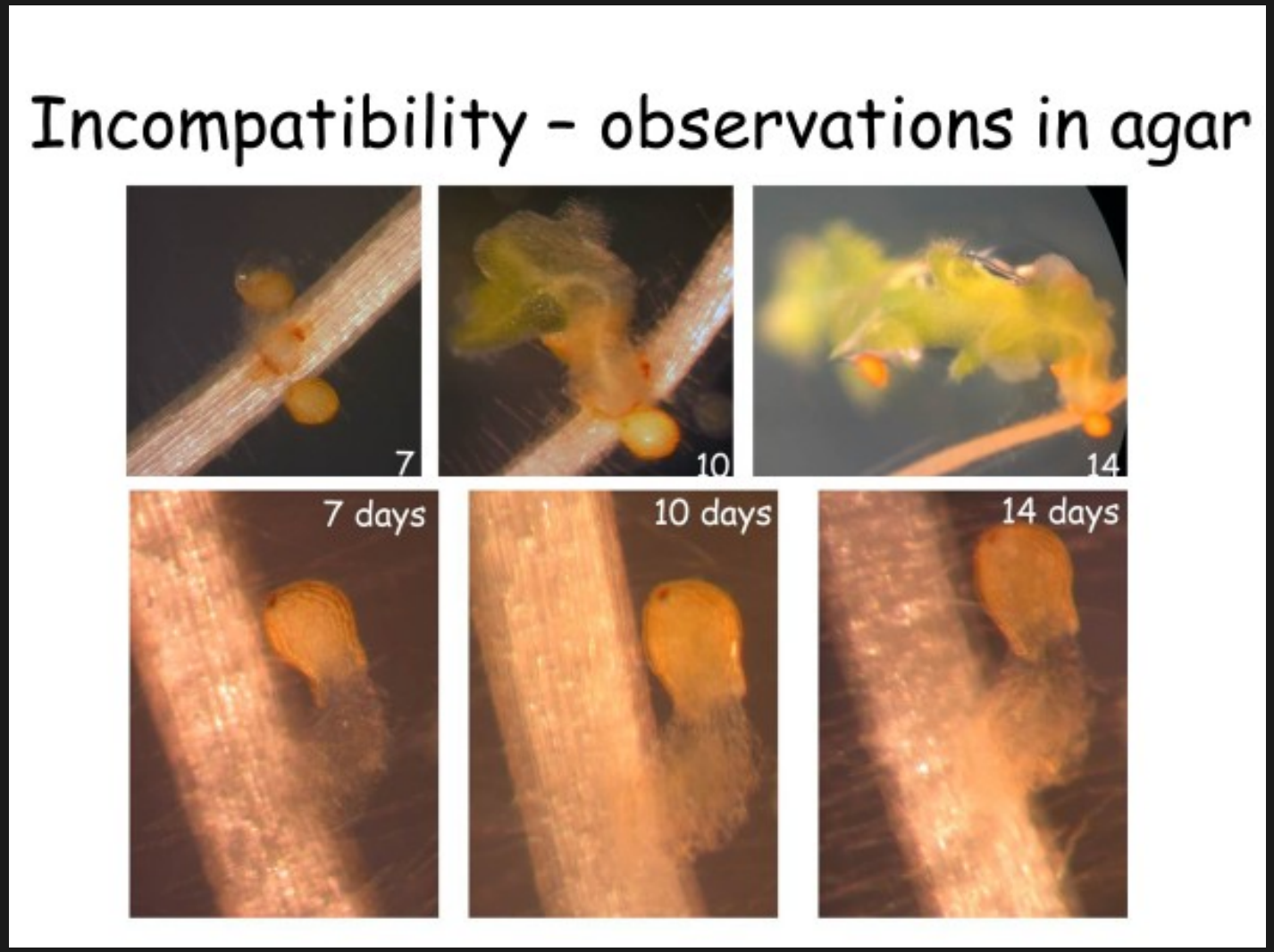
# What is known about *Striga hermonthica?*




- There are several potential interventions for sorghum hosts to the *Striga* lifecycle
- Low *Striga* germination stimulant activity of root exudates already discovered (*lgs1*) and being exploited for sorghum resistance (Gobena et al. 2017)
- Incompatibility is another potential *Striga* resistance mechanism in sorghum
- Multiple barriers to parasitism results in more stable *Striga* resistance

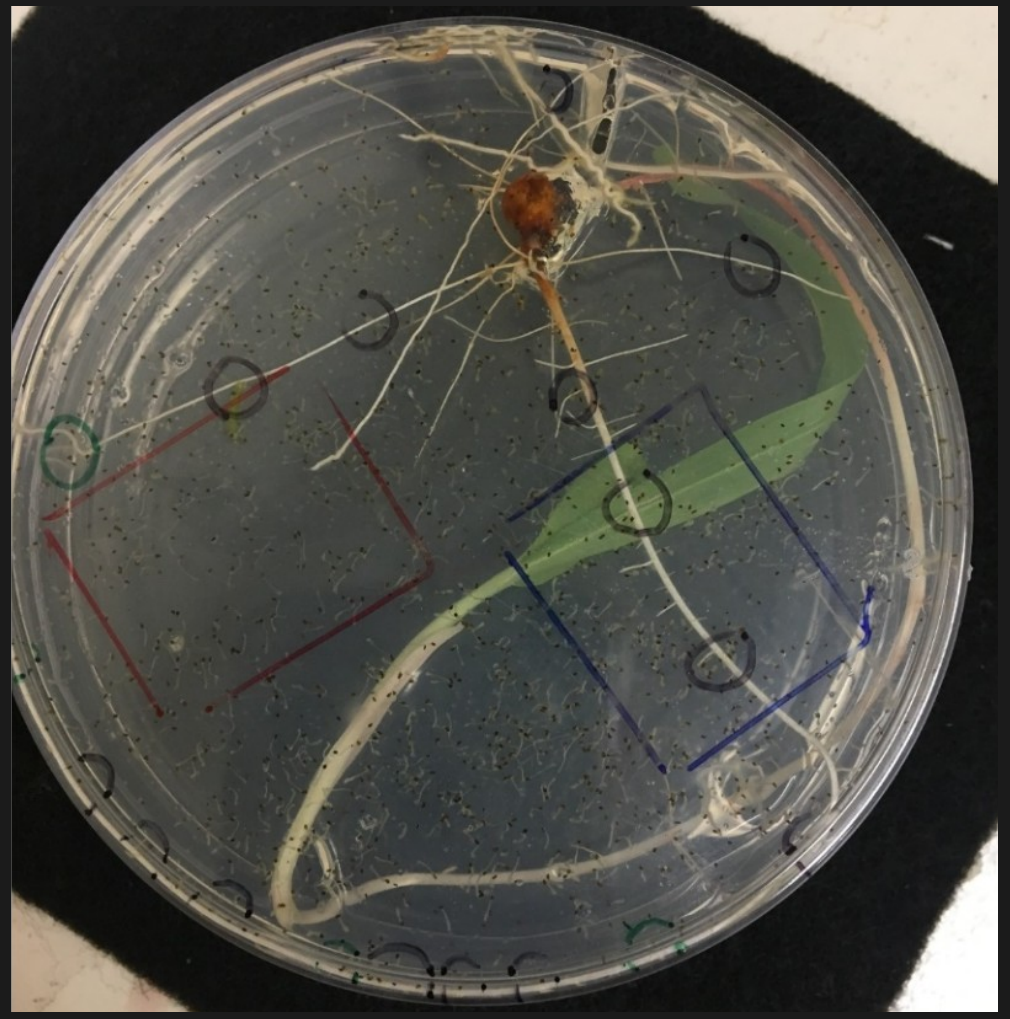
# Goals of the research project

- Phenotype a population of 58 recombinant inbred lines from a resistant donor sorghum having both *lgs1* and post-attachment *Striga* resistance (incompatibility) to be used in combination with whole genome resequencing of pooled DNA from contrasting phenotypes (Bulk Segregant Analysis) in hopes of discovering gene(s) involved in post-attachment *Striga* resistance



# Materials and methods

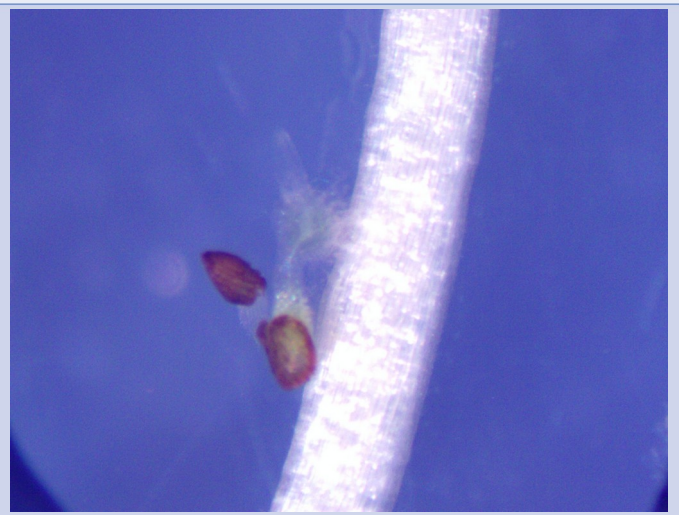
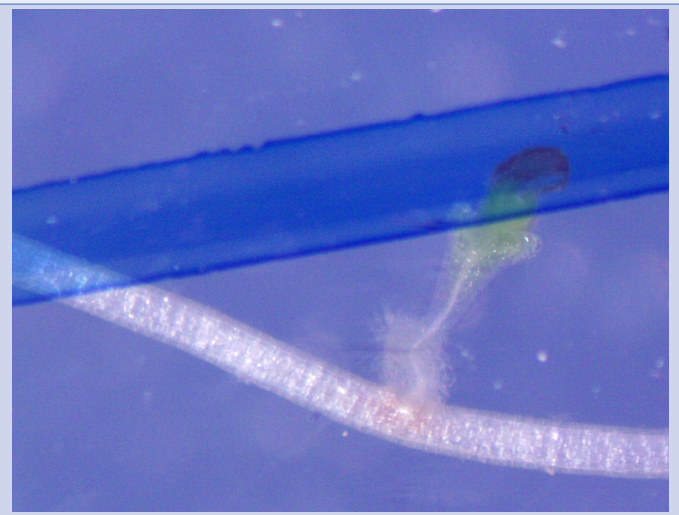
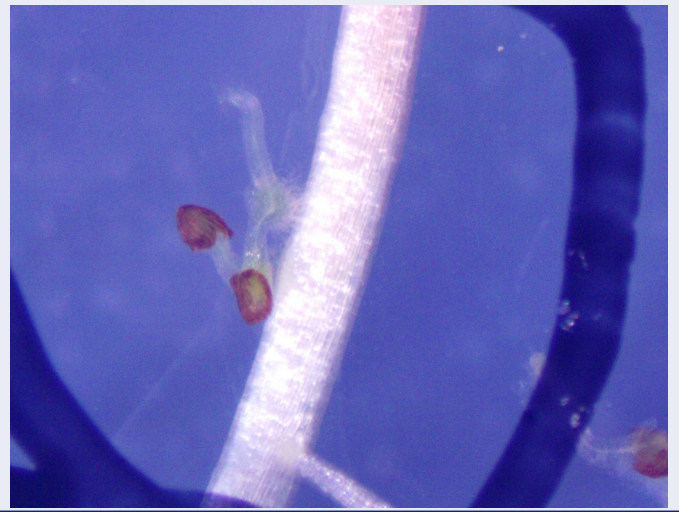
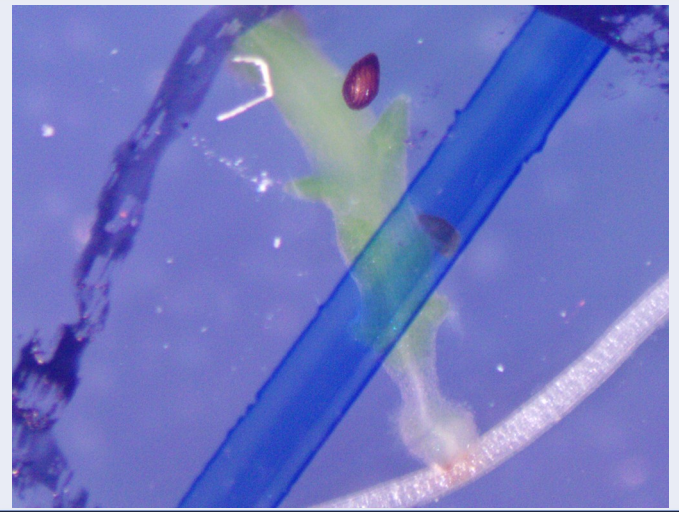
- *Striga* and sorghum seeds were surfaced sterilized
- *Striga* was conditioned for 14 days before embedding in agar
- A germinated sorghum seed was then planted into the agar containing conditioned *Striga*
- After a few days of co-culture plates were sprayed with GR24, a synthetic strigolactone, to overcome low *Striga* germination stimulant activity among progeny lines
- Attachments of *Striga* on sorghum roots were marked and numbered after 10 days of co-culture.



# Materials and methods (continued)

- At 14 days, individual attachments on each plate were counted and imaged
- At 21 days, plates were reimaged to analyze for growth of individual *Striga* attachments on each sorghum line
- Individuals were quantified for attachment growth percentage as a phenotype to use for determining bulk segregant members

# Analysis

| | Attachments that did not grow | Attachments that did grow |
|---|---|---|
| Attachments at 14 days |  |  |
| Attachments at 21 days |  |  |

# Conclusion

- Our screening of 58 inbred lines from the cross of two parents did not yield a large enough separation of individuals for post-attachment *Striga* resistance
- The lowest was 0% post-attachments success of *Striga* and the highest was 67% but there were not enough individuals in each pool to perform bulk segregant analysis
- The process did confirm *LGS1* is not responsible for post-attachment resistance
- It also appears that post-attachment resistance is conferred from both parents, not only the donor of *lgs1*

| DESG. | PEDIGREE | LGS1 allele | Avg. no. attachments/plant @14 | Avg. no. attachments that grew between 14 and 21 days | Percent attached that grew | Total no. attachments that grew between 14 and 21 days |
|---|---|---|---|---|---|---|
| PSL985003; P9403 | (SRN39 × P954063)F16 | | 3.3 | 0.0 | 0% | 0 |
| PSL985006 | (SRN39 × P954063)F16 | | 2.5 | 0.0 | 0% | 0 |
| PSL985009; P9406 | (SRN39 × P954063)F16 | | 3.3 | 0.0 | 0% | 0 |
| PSL985027 | (SRN39 × P954063)F15 | | 2.0 | 0.0 | 0% | 0 |
| PSL985028 | (SRN39 × P954063)F15 | | 2.0 | 0.0 | 0% | 0 |
| PSL985036 | (SRN39 × P954063)F15 | | 7.0 | 0.0 | 0% | 0 |
| PSL985039 | (SRN39 × P954063)F15 | lgs1-1 | 1.7 | 0.0 | 0% | 0 |
| PSL985042 | (SRN39 × P954063)F15 | | 4.0 | 0.0 | 0% | 0 |
| PSL985051 | (SRN39 × P954063)F15 | | 5.0 | 0.0 | 0% | 0 |
| PSL985053 | (SRN39 × P954063)F15 | | 2.0 | 0.0 | 0% | 0 |
| PSL985058 | (SRN39 × P954063)F15 | | 4.0 | 0.0 | 0% | 0 |
| PSL985004 | (SRN39 × P954063)F16 | | 6.7 | 0.3 | 5% | 1 |
| PSL984995; P9407 | (SRN39 × P954063)F16 | | 3.0 | 1.0 | 33% | 2 |
| PSL985031 | (SRN39 × P954063)F15 | | 3.0 | 1.0 | 33% | 3 |
| PSL985032 | (SRN39 × P954063)F15 | | 3.0 | 1.0 | 33% | 2 |
| PSL985037 | (SRN39 × P954063)F15 | | 3.0 | 1.0 | 33% | 1 |
| PSL985429 | SRN39 | lgs1-1 | 9.0 | 3.0 | 33% | 3 |
| PSL985026 | (SRN39 × P954063)F15 | | 2.0 | 1.0 | 50% | 1 |
| PSL985035 | (SRN39 × P954063)F15 | | 4.0 | 2.0 | 50% | 2 |
| PSL985060 | (SRN39 × P954063)F15 | | 3.0 | 1.5 | 50% | 3 |
| PSL984997 | (SRN39 × P954063)F16 | | 3.0 | 2.0 | 67% | 4 |

# Future Directions

- We plan on screening a much larger population from more contrasting parents in the future
- This screening will be performed in hopes of identifying two distinct groups that either have or do not have post-attachment *Striga* resistance
- After identifying two distinct groups, bulk segregant analysis will be performed to hopefully identify the gene(s) responsible for post-attachment *Striga* resistance
- Combined DNA  from each bulk segregant pool will be whole genome sequenced and consistent differences between pools will point to candidate genes which will then be verified in other sorghum lines/populations

# Works Cited


- Gobena D, Shimels M, Rich PJ, Ruyter-Spira C, Bouwmeester H, Kanuganti S, Mengiste T, Ejeta G. Mutation in sorghum *LOW GERMINATION STIMULANT 1* alters strigolactones and causes *Striga* resistance. PNAS. 2017 Apr 25;114(17):4471-6.

- Runo S, Kuria EK. Habits of a highly successful cereal killer, *Striga*. PLoS Pathog. 2018 Jan 11;14(1):e1006731. doi: 10.1371/journal.ppat.1006731. PMID: 29324906; PMCID: PMC5764402.